

MVC (Model-View-Controller) Test

Section 1: Basic Concepts (5 Questions)

1. What is the primary purpose of the Model in MVC?

- A) Handle user input
- B) Store and manage data
- C) Display data to users
- D) None of the above

Answer: B

2. In the MVC pattern, what does the Controller do?

- A) Handles the presentation logic
- B) Manages user input and updates the Model
- C) Stores data in the database
- D) Controls the layout of the View

Answer: B

3. Which component in MVC is responsible for rendering the user interface?

- A) Model
- B) View
- C) Controller
- D) Database

Answer: B

4. How does the View receive data in an MVC application?

- A) Directly from the database
- B) From the Controller
- C) By making API calls
- D) Through a background service

Answer: B

5. Which of the following best describes the relationship between Model and View?

- A) The Model updates the View directly
- B) The View fetches data directly from the Model
- C) The View updates the Model
- D) The Controller updates the View with data from the Model

Answer: D

Section 2: ASP.NET MVC Framework (5 Questions)

6. In ASP.NET MVC, what is the purpose of the `ActionResult` return type?
- A) It only returns JSON data
 - B) It represents the result of an action method
 - C) It is used only for error handling
 - D) It is a replacement for ViewModels

Answer: B

7. Which routing format is typically used in ASP.NET MVC?
- A) `/Controller/Action/ID`
 - B) `/View/Controller/ID`
 - C) `/Model/View/Controller`
 - D) `/Action/Controller/ID`

Answer: A

8. What is the purpose of the `ViewBag` in ASP.NET MVC?
- A) It allows passing data between the Controller and View
 - B) It is used to store session data
 - C) It is used for dependency injection
 - D) It manages database connections

Answer: A

9. What is the default return type of an MVC Controller action method?
- A) JSON
 - B) `ViewResult`
 - C) `XmlResult`
 - D) String

Answer: B

10. In ASP.NET MVC, what is `TempData` primarily used for?
- A) Passing data from one request to another
 - B) Storing session state
 - C) Caching data in the browser
 - D) Logging errors

Answer: A

Section 3: Advanced Concepts (5 Questions)

11. What is the purpose of a ViewModel in MVC?
- A) It acts as a database schema
 - B) It combines multiple Models for a View
 - C) It replaces the Controller
 - D) It controls routing

Answer: B

12. What is Dependency Injection in the context of ASP.NET MVC?
- A) A way to pass data between Views
 - B) A method to dynamically load controllers
 - C) A design pattern for managing dependencies
 - D) A way to validate user input

Answer: C

13. What is the main advantage of using Partial Views?
- A) Improves performance by caching Views
 - B) Allows reuse of UI components
 - C) Reduces the need for Controllers
 - D) Avoids using JavaScript

Answer: B

14. What is the purpose of `JsonResult` in an MVC Controller?
- A) It renders HTML views
 - B) It allows returning JSON-formatted data
 - C) It retrieves data from the database
 - D) It handles error pages

Answer: B

15. How do you specify that an MVC action should only be accessed via HTTP POST?
- A) `[HttpGet]` attribute
 - B) `[PostOnly]` attribute
 - C) `[HttpPost]` attribute
 - D) `[HttpAllow]` attribute

Answer: C

Section 4: Security & Authentication (5 Questions)

16. What is the purpose of Identity in ASP.NET MVC?
- A) Managing user authentication and authorization
 - B) Encrypting database records
 - C) Logging error messages
 - D) Enforcing MVC architecture

Answer: A

17. Which of the following is a secure way to store passwords in ASP.NET MVC?
- A) Storing as plain text
 - B) Using MD5 hashing
 - C) Using password hashing with salt
 - D) Encrypting with a hardcoded key

Answer: C

18. What does the `[Authorize]` attribute do in ASP.NET MVC?
- A) Restricts access to authenticated users
 - B) Logs all user requests
 - C) Validates user input
 - D) Encrypts sensitive data

Answer: A

19. How can you prevent Cross-Site Request Forgery (CSRF) attacks in ASP.NET MVC?
- A) Use the `[ValidateAntiForgeryToken]` attribute
 - B) Disable cookies
 - C) Encrypt all form inputs
 - D) Only allow GET requests

Answer: A

20. What is the purpose of Role-Based Authentication in ASP.NET MVC?
- A) Allowing only certain users to access specific features
 - B) Encrypting user passwords
 - C) Managing database transactions
 - D) Optimizing query performance

Answer: A

Task: [Create a Simple MVC Action and View](#)

Objective: Implement a basic ASP.NET MVC Controller action and its corresponding View.

Instructions:

1. **Create a new Action Method** in an existing or new **Controller** named `HomeController`.
 - o Add an action method named `GreetUser` that returns a **View**.
 - o Pass a string message, e.g., "Welcome to MVC!", to the **View** using `ViewBag`.

```
public class HomeController : Controller
{
    public ActionResult GreetUser()
    {
        ViewBag.Message = "Welcome to MVC!";

        return View();
    }
}
```

2. **Create a View** for the `GreetUser` action.

- Inside the `Views/Home/` folder, create a new **View** named `GreetUser.cshtml`.
- Display the message passed from the Controller.

```
@* Views/Home/GreetUser.cshtml *@
@model string
<h2>@ViewBag.Message</h2>
```

3. **Run the project** and navigate to `/Home/GreetUser` to see the message displayed.

When students visit `/Home/GreetUser`, they should see "**Welcome to MVC!**" displayed on the page.

This exercise tests their ability to create a **Controller action**, pass data using `ViewBag`, and render it in a **View**. It's simple but ensures they understand the **fundamental MVC flow**.